

Date of Assignment: 6 Aug 2025	Full Marks 90
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Answer all questions. Symbols have their usual meanings. Make suitable assumptions wherever necessary. Please state your assumptions clearly. You may refer to your class notes.
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Question 1 (15 points)

Air at 20°C and relative humidity 35% is blown by a fan through a 20 m long duct of cross-section 22 cm X 22 cm. The average velocity in the duct is 5 m/s. The horizontal duct has two 90° bends each having loss coefficient $K_{\text{loss}}=1.2$ and the duct surface roughness is $\epsilon=0.00015$ m. The density and dynamic viscosity of air are $\rho=1.2$ kg/m³ and $\mu=1.8 \times 10^{-5}$ Pas.

- (a) Find the volume flow rate in m³/s. (2)
- (b) Find the hydraulic diameter of the square duct and Reynolds number for this flow. (3)
- (c) What is the friction factor. Refer to the Moody's chart (attached). (5)
- (d) Find the fan pressure head in Pa required to sustain the air flow in the duct. (5)

Question 2 (10 points)

Air at 40°C dry-bulb temperature and 25°C wet bulb temperature is passed through a insulated water spray chamber. The temperature of the outlet air is 30°C.

- (a) Draw the process in a psychrometric chart and find the amount of moisture in g added per kg of dry air. (5)
- (b) What is the efficiency of the process? (5)

Question 3 (15 points)

The air-conditioned space is to be maintained at 25°C dry bulb temperature and 50% relative humidity *i.e.*, the room outlet condition. The sensible cooling load in the space during summer is 45kW and latent heat load is 20kW *i.e.*, the amount of heat generation in the room. What is the ratio of latent load to total load? Draw the load-ratio line in the psychrometric chart.

Question 4 (20 points)

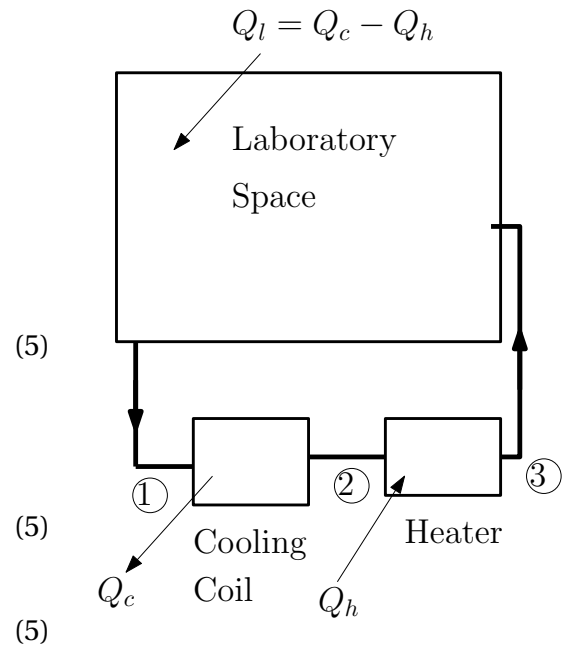
In an air-conditioning unit 3.5 m³/s of air at 27°C dry-bulb temperature, 50% relative humidity enters at standard atmosphere pressure. The leaving condition of air is 13°C dry-bulb temperature and 90% relative humidity. Using properties from the psychrometric chart

- (a) Find dew point of the air at 27°C dry-bulb temperature, 50% relative humidity (5)
- (b) Calculate the rate of energy addition or removal in kW inside the room. (5)
- (c) Find the ratio of latent heat to total heat. (5)
- (d) Find the rate of moisture addition or removal from air in g/s inside the room (5)

Question 5 (15 points)

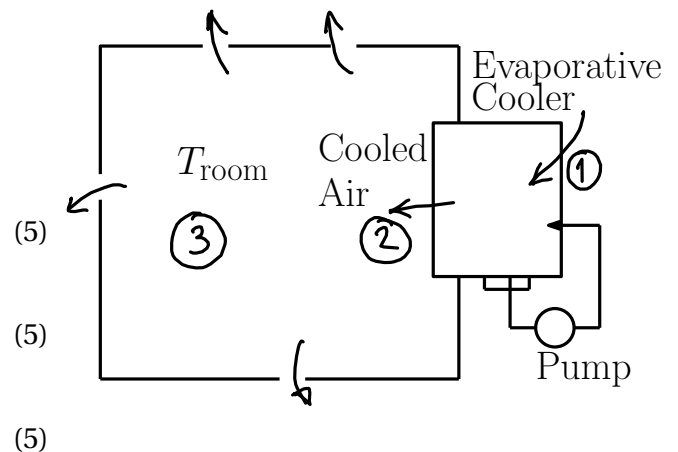
A laboratory space to be maintained at 24°C dry-bulb temperature and 60% relative humidity (*i.e.*, point 1) experiences a heating load consisting of sensible load of 24kW and a latent load of 36kW. The closed circuit air-conditioning system is equipped for reheating the air leaving the cooling coil. The cooling coil is selected to provide outlet air at 9°C dry-bulb-temperature and 95% relative humidity (*i.e.*, point 2).

- Draw the load-ratio line and the cooling and reheating process on a psychrometric chart (dry-bulb temperature versus humidity ratio). Show the points 1, 2 and 3 in the chart.
- Find the dry-bulb temperature of air supplied to the room in °C.
- Find the mass flow rate of air in kg/s

**Question 6** (15 points)

In an evaporative cooler, 0.6 kg/s of air at 101.325 kPa, 42°C (DBT) and 18°C (WBT) (*i.e.*, point 1) is cooled and humidified. The room temperature is maintained at 27°C (*i.e.*, point 3). The efficiency of the evaporative cooler is 90%.

- draw the process 1-2-3 in the psychrometric chart
- find the required make-up water flow rate in kg/h
- find sensible cooling capacity of the evaporative cooler in kW (*i.e.*, assume only sensible energy load in the room).



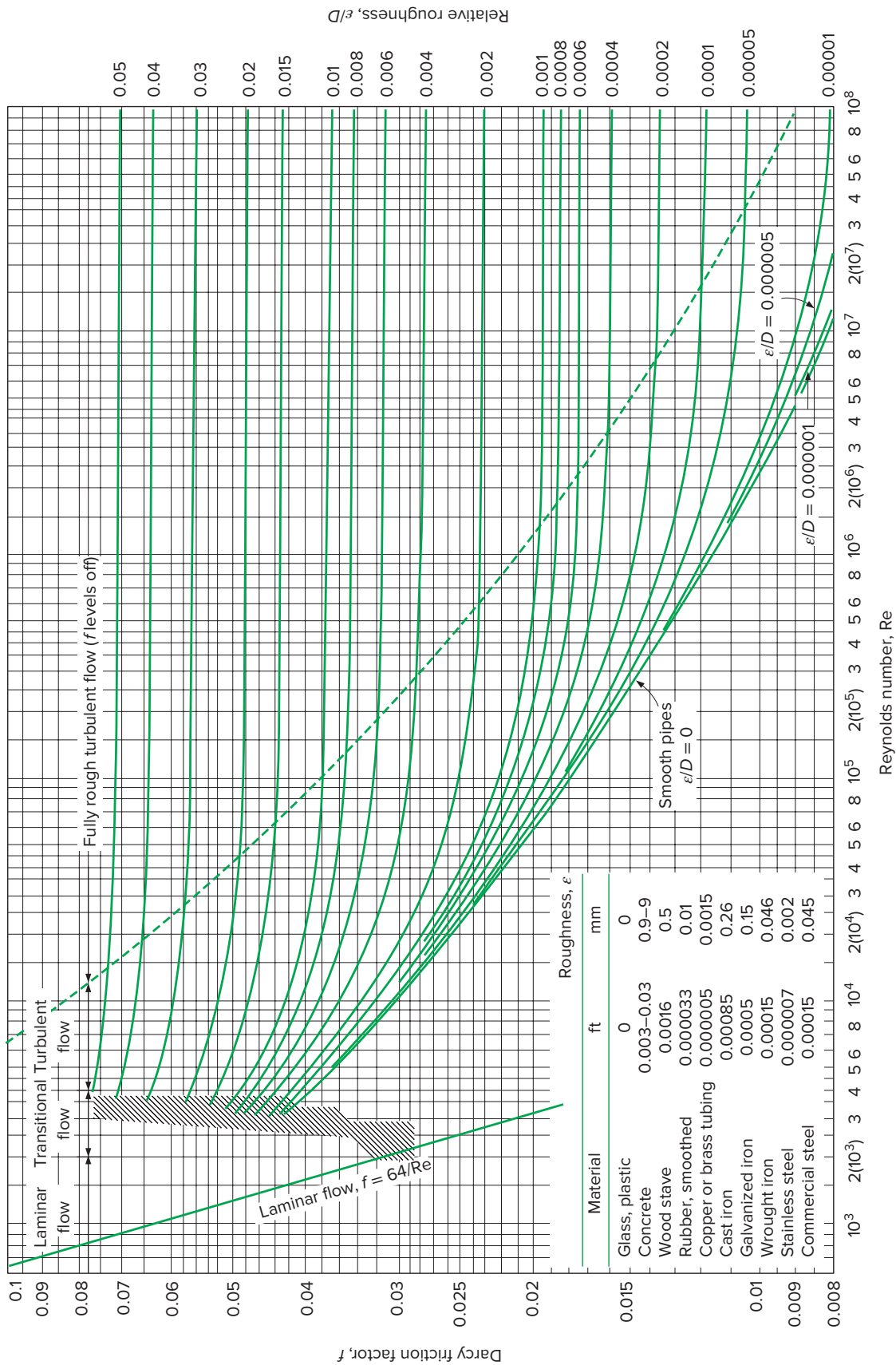


FIGURE A-27

The Moody chart for the friction factor for fully developed flow in circular pipes for use in the head loss relation $\Delta P_L = f \frac{L}{D} \frac{\rho V^2}{2}$. Friction factors in the turbulent flow are evaluated from the Colebrook equation $\frac{1}{\sqrt{f}} = -2 \log_{10} \left(\frac{\epsilon/D}{3.7} + \frac{2.51}{Re \sqrt{f}} \right)$.

